

KB-005-CCS-Charging-Guide

KB-005 — CCS Charging — Full Driver Guide & Troubleshooting

Category: CCS Connector — Education & Troubleshooting
Call Volume: ~12% of contacts
Audience: Both new EV drivers unfamiliar with CCS and experienced drivers hitting CCS-specific faults
Primary Remote Actions: Remote reset, status check, technician dispatch

Why This Matters

CCS (Combined Charging System) is the dominant fast-charging standard in North America and Europe, used by most non-Tesla EVs. However, its two-part connector design, handshake complexity, and physical locking mechanism generate confusion and specific failure modes not seen with AC charging. Agents must be fluent in both explaining CCS to new users and diagnosing CCS-specific technical failures.

Background: What Is CCS?

CCS Standards

Standard	Also called	Used in	Connector description
CCS1 (CCS Type 1)	SAE J1772 Combo	North America	J1772 AC pins on top + 2 large DC pins on bottom
CCS2 (CCS Type 2)	IEC 62196 Combo	Europe, some international	Mennekes AC pins on top + 2 large DC pins on bottom

The top portion (AC pins) handles Level 2 AC charging. The bottom portion (DC pins) handles DC fast charging (DCFC). Both are part of a single combined connector — hence “Combined Charging System.”

Power Levels

Level	Power range	Approximate charge speed
Level 2 AC (CCS AC portion)	7–22 kW	Adds ~20–60 miles per hour
DC Fast Charging (CCS DC portion)	50–350 kW	Adds ~100–300+ miles per 20–30 min

How to Identify a CCS Charger

- DC Fast Charger with a large, two-tier connector (top half smaller, bottom half with 2 large round pins)
- Often labeled “CCS,” “DC Fast,” or “DCFC” on the station
- Typically tethered (cable attached to charger, not a loose socket)
- Usually has a large rectangular display showing session status

Which Vehicles Use CCS

Most non-Tesla EVs sold in North America and Europe use CCS1 (NA) or CCS2 (EU), including: - BMW, Chevrolet, Ford, Hyundai, Kia, Nissan (newer models), Polestar, Rivian, Stellantis brands (Jeep, RAM), Volkswagen Group (Audi, Porsche, VW)

Note: Tesla vehicles use NACS (North American Charging Standard) natively, though many newer Teslas include a CCS adapter or NACS-to-CCS compatibility. CHAdeMO (used by older Nissan LEAF and Mitsubishi) is a separate standard entirely and is not CCS.

Section A — How to Start a CCS Charging Session (Driver Education)

Many calls come from drivers who are new to CCS and simply don’t know the correct process. Walk them through this step by step.

Step-by-Step CCS Session Start

Step 1 — Check vehicle compatibility > “First — does your car have a CCS port? It’s on the outside of the vehicle and looks like a standard J1772 socket on top with two larger pins below it. If you’re not sure, check your owner’s manual or look up your vehicle model.”

Step 2 — Pull up to the charger Park close enough that the tethered cable reaches the vehicle inlet without strain. Overstretched cables can damage the connector and the inlet.

Step 3 — Power down the vehicle (if required) Most modern EVs allow charging while the car is powered on, but some manufacturers require the vehicle to be in Park and powered down (key off or in ready mode). Check vehicle-specific guidance: - BMW, Hyundai, Kia: typically fine in “ready” mode - Some older EVs: may require powered down

Step 4 — Locate the charging port on the vehicle The CCS inlet is typically on the driver’s side rear quarter panel, front quarter panel, or rear center (varies by manufacturer). Open the port cover — this may be a mechanical flap, electronic button, or automatic (opens when a compatible connector is detected).

Step 5 — Prepare the connector Pick up the CCS connector. It is heavy compared to an AC plug. Hold it by the handle — not the cable. Align the connector with the inlet, paying attention to orientation (the connector is keyed and will only fit one way).

Step 6 — Insert the connector Insert firmly until you feel and/or hear a click. The latch should engage. Do not force it. If it doesn’t click: - Check alignment — the connector may be slightly rotated - Inspect for debris in the vehicle inlet - Try a slightly different angle

Step 7 — Authorize the session Authorization methods vary by station: - **App:** Open the network app, tap Start Session, select the station - **RFID card:** Tap card against the RFID reader on the charger (usually marked with a contactless payment icon) - **Tap-to-pay:** Tap credit or debit card against the card reader - **QR code:** Scan the QR code on the station with your phone to open a guest checkout flow - **Plug-and-charge (ISO 15118):** Some vehicles and networks support automatic authentication when the cable is plugged in — no additional action needed

Step 8 — Confirm session started - The charger screen should show “Charging” or a session in progress - The vehicle’s dashboard or instrument cluster should show a charging indicator - You may hear a relay click as the contactors close and power begins flowing - The connector latch will lock — you will not be able to remove the cable while the session is active

Step 9 — During the session - The vehicle can be locked (and should be, for security) - You can monitor progress in the app or on the charger screen - Do not kick, pull, or force the cable — it is mechanically locked

Step 10 — Ending the session - Tap “Stop Session” in the app OR on the charger screen - Wait for the screen to confirm the session is complete - The connector latch will release - Remove the connector and return it to the holder - Close the vehicle port cover

Section B — CCS-Specific Troubleshooting

Problem: Connector Won’t Insert

“Can you describe what’s happening — is it not going in at all, or going in partway?”

Not going in at all: 1. Alignment issue — rotate slightly and re-try. The connector is keyed. 2. Debris in vehicle inlet — look inside with a flashlight. Do not insert the connector if debris is present. 3. Wrong connector type — confirm the vehicle uses CCS. CHAdeMO and NACS connectors look different. 4. Physical damage to connector — if pins are visibly bent, do not insert. Dispatch a technician.

Going in partway but not clicking: 1. The vehicle inlet cover may not be fully open — check that the port door is completely retracted. 2. The connector may need a firmer push — CCS connectors require more insertion force than AC plugs. 3. Inlet damage — if the vehicle’s CCS port has debris, corrosion, or damage, do not force it.

Problem: Session Won’t Start After Connector Is Inserted

Follow the diagnostic in KB-001 (Session Won’t Start), with these CCS-specific additions:

CCS handshake failure: The CCS DC charging handshake (ISO 15118 / DIN 70121 / IEC 61851) is more complex than AC charging. It involves: - Control Pilot (CP) signal negotiation - Proximity Pilot (PP) confirmation that the connector is properly seated - High-level communication over PLC (Power Line Communication) for ISO 15118

If the handshake fails: 1. Try unplugging and reinserting firmly — PP signal depends on a complete mechanical connection 2. Soft reset the charger 3. Ask if the driver has tried a different CCS charger recently — if all CCS fast chargers fail, the vehicle’s PLC controller may have a fault

Charger screen shows “Vehicle not responding” or “Communication error”: Vehicle-side pilot signal or PLC communication fault. See KB-003. This is vehicle-side.

Charger screen shows “Connector not detected”: The PP (Proximity Pilot) circuit isn’t registering a seated connector. 1. Unplug and re-seat firmly 2. Check connector and inlet for debris 3. If the connector latch mechanism is worn, it may not be generating the PP signal — dispatch a technician

Problem: DC Fast Charging Won't Start but Level 2 (AC) Works

This pattern — DC fails, AC succeeds at the same vehicle inlet — points to:

1. **Vehicle DC inlet fault** — the DC pins (lower portion of CCS) may be damaged while AC pins (upper) are fine 2. **Vehicle BMS or DC fast charge module fault** — the vehicle's DC fast charge pathway has a software or hardware issue 3. **This is almost always vehicle-side**

“If your car charges fine on a regular Level 2 charger but DC fast charging consistently fails, the issue is with your car's DC fast charging system rather than the stations. I'd recommend a visit to your dealership — they can run a specific diagnostic on the DC charging pathway.”

Problem: Charging Stops Mid-Session on CCS

Refer to KB-001 Section B (Session Ended Early), with these CCS-specific notes:

Thermal derating on DC fast charging: DC fast charging generates significant heat in the vehicle battery, the charger power module, and the cable itself. Thermal management systems on both sides will reduce power or stop the session if temperatures exceed safe limits. This is normal and expected, especially: - In hot weather (ambient above 30°C / 86°F) - After a previous fast charge session (battery already warm) - At high SoC (above 80%)

Communication dropout mid-session: If the PLC communication link drops during a session (due to charger modem issues, electrical interference, or a loose connector), the charger will stop power delivery as a safety measure. 1. Re-seat the connector 2. Soft reset and retry

Vehicle charge limit reached: Many EVs are factory-set to a charging limit (often 80% or 90%) to protect battery longevity. The driver may not be aware.

“It's worth checking your vehicle's charging settings — most EVs let you set a maximum charge level. If it's set to 80%, the session will end at 80% even if you expected it to go higher.”

Problem: Connector Stuck in Vehicle

This is covered fully in KB-008 (Stuck Cable), but CCS-specific points:

- The CCS connector locks mechanically when a session is active. It will not release until the session officially ends.
- If the session has ended on the screen but the connector is still locked: send a remote connector unlock command.
- Some vehicle models have a manual emergency cable release inside the vehicle (typically in the trunk or under a panel). Check vehicle documentation.

- Do not advise the driver to pull forcefully — this can damage the vehicle’s CCS inlet, which is an expensive repair.

Section C — CCS vs. Other Connector Standards

Agents must be able to quickly identify if a driver has the wrong connector type, as this is a common source of confusion.

Standard	Connector look	Common vehicles	Compatible with CCS charger?
CCS1	J1772 top + 2 DC pins bottom	Most US non-Tesla EVs	Yes — this is CCS
CCS2	Mennekes top + 2 DC pins bottom	EU EVs	Yes — this is CCS (EU variant)
CHAdE MO	Round, single-piece DC connector, separate from AC	Older Nissan LEAF, Mitsubishi	No — needs CHAdE MO port
NACS (Tesla)	Slim, single-piece oval connector	Tesla, some new Ford/GM/Rivian	Adapter required for CCS
J1772 only	Standard AC plug, no DC pins	All EVs (AC charging only)	Only for Level 2, not DCFC
GB/T	Chinese standard	Chinese-market EVs	No

If a driver with a CHAdE MO vehicle is at a CCS-only fast charger: > “Unfortunately, CHAdE MO and CCS are different connector standards and aren’t compatible with each other. That station only has CCS ports. Do you want me to help you find the nearest station that has a CHAdE MO connector?”

If a Tesla driver is at a CCS station: > “Tesla’s native connector is NACS. Some CCS stations have a NACS port or a CCS-to-NACS adapter available — does this station have one? If not, you’d need an adapter or a Tesla Supercharger.”

Section D — CCS Station Identification Guide (For Agents)

When a driver can't find the station ID, agents can help identify the charger type from the driver's description:

Driver describes	Likely charger type
"Big box on a pole, thick cable, two big pins at the bottom"	DC Fast Charger with CCS
"Small box on a wall, looks like a gas pump nozzle"	Level 2 AC charger (J1772)
"Round connector, single piece"	CHAdEMO
"Slim, one-piece connector, Tesla logo"	Tesla Supercharger / NACS
"Two connectors hanging off the same unit"	Dual-port DCFC (often CCS + CHAdEMO)

Escalation Criteria

Condition	Action
DC fails, AC works (same inlet)	Vehicle-side — direct to manufacturer
CCS connector has bent or damaged pins	Dispatch technician; do not allow use
Connector locked and won't release after session end	Remote unlock; if fails, dispatch
Repeated CCS handshake failures across multiple stations	Vehicle BMS/PLC issue — direct to dealer
Driver reports sparks or burning from CCS connector	KB-009 Safety Protocols immediately

Agent Reminders

- CCS connectors are heavier and require more insertion force than AC plugs — this surprises new users.
- If DC fast charging fails but AC works, it is almost always the vehicle, not the charger.
- The connector locks while charging. Trying to remove it during a session is normal driver behavior — calmly explain they need to end the session first.
- CHAdEMO and CCS are not compatible. Know the difference and have a plan for CHAdEMO drivers at CCS-only stations.

- “Plug-and-charge” (ISO 15118) is increasingly common — some vehicles start the session automatically when plugged in. If a driver says “I don’t know how to start it,” ask if their car does plug-and-charge first.